HOME LAB NETWORK
FIREWALL RULES DOCUMENTATION

pfSense 2.8.1-RELEASE | Protectli FW6E | Mullvad VPN over WireGuard

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1. Firewall Architecture Overview

This document details every firewall rule configured on pfSense 2.8.1-RELEASE running on a Protectli FW6E appliance. All internet-bound traffic is routed through Mullvad VPN (WireGuard) with automatic failover between two geographic endpoints. A VPN kill switch ensures no traffic leaks to the ISP if both tunnels fail.

Design Principles:

- Default deny — All traffic is blocked unless explicitly permitted by a rule
- VPN-forced routing — All VLAN traffic exits through VPN_FAILOVER gateway group (except VLAN50_MGMT)
- IPv6 disabled — Blocked on every interface to prevent tunnel bypass
- DNS lockdown — DNS resolves only through pfSense via VPN; direct DNS to ISP and encrypted DNS (DoH/DoT) are blocked
- Inter-VLAN isolation — RFC1918 block prevents lateral movement between VLANs
- Management exception — VLAN50_MGMT has no inter-VLAN block and no VPN, enabling direct management of all network segments

Interfaces / Interface Assignments

Interface Assignments | Interface Groups | Wireless | VLANs | QinQs | PPPs | GREs | GIFs | Bridges | LAGGs

Interface	Network port	
WAN	igb0 [redacted]	
LAN	igb1 [redacted]	Delete
RESERVED_OOB	igb2 [redacted]	Delete
RESERVED_LAB	igb3 [redacted]	Delete
RESERVED_HA	igb4 [redacted]	Delete
RESERVED_EXP	igb5 [redacted]	Delete
VPN_CHI	tun_wg0 (tun_wg0)	Delete
VLAN10_USERS	VLAN 10 on igb1 - lan (VLAN10_USERS)	Delete
VLAN20_IOT	VLAN 20 on igb1 - lan (VLAN20_IOT)	Delete
VLAN30_GUEST	VLAN 30 on igb1 - lan (VLAN30_GUEST)	Delete
VLAN40_LAB	VLAN 40 on igb1 - lan (VLAN40_LAB)	Delete
VLAN50_MGMT	VLAN 50 on igb1 - lan (VLAN50_MGMT)	Delete
VPN_NYC	tun_wg1 (tun_wg1)	Delete
Tailscale_Int	tailscale0 (tailscale0)	Delete

Save

Firewall > Rules > Interface tabs overview showing all configured interfaces

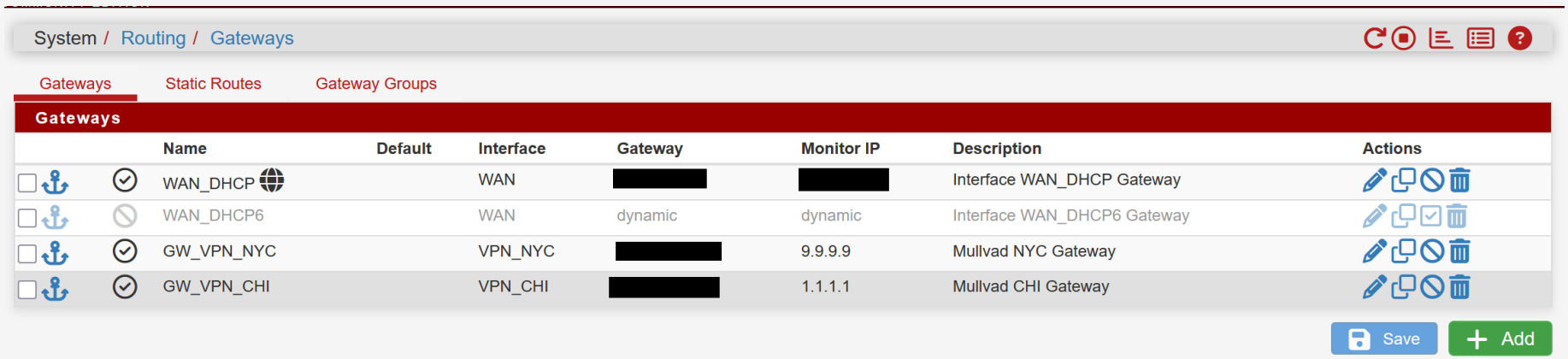
2. Gateway Configuration

All VPN-routed traffic uses the VPN_FAILOVER gateway group, which provides automatic failover between Chicago (primary) and New York City (secondary) Mullvad endpoints.

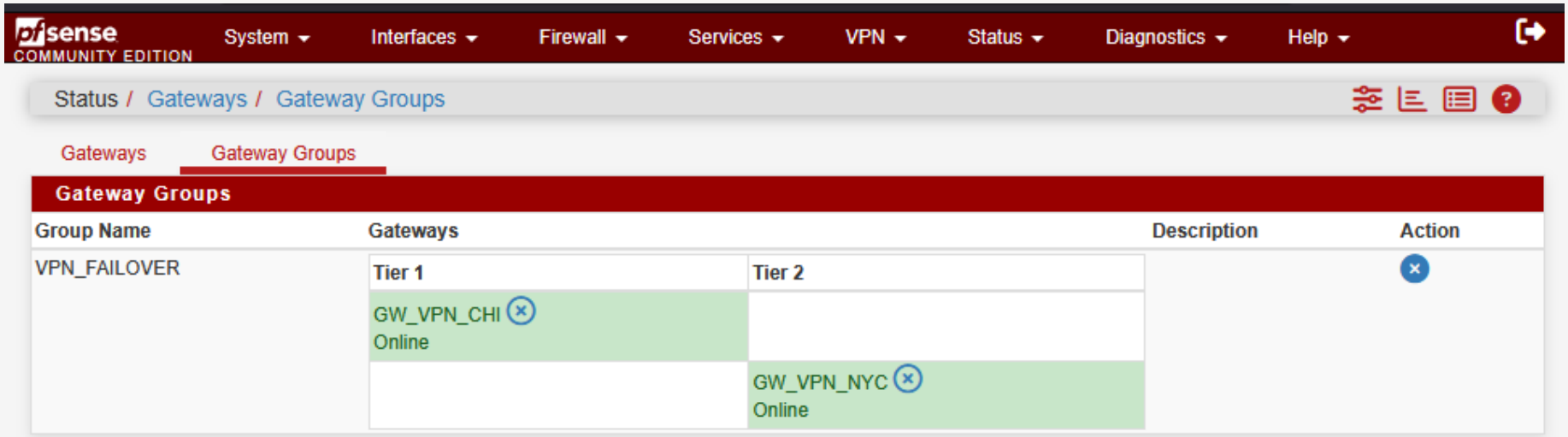
2.1 Gateway Summary

Gateway Name	Interface	Monitor IP	Tier	Role	Description
GW_VPN_CHI	VPN_CHI (tun_wg0)	1.1.1.1	Tier 1 (Primary)	Primary VPN exit	Mullvad Chicago Gateway
GW_VPN_NYC	VPN_NYC (tun_wg1)	9.9.9.9	Tier 2 (Failover)	Failover VPN exit	Mullvad NYC Gateway
VPN_FAILOVER	Gateway Group	N/A	CHI=T1 / NYC=T2	Failover group	Auto-failover: CHI primary, NYC backup
WAN_DHCP	WAN (igb0)	Auto	N/A	ISP uplink	ISP DHCP gateway (kill switch blocks this for VLANs)

Rationale: Geographic separation between Chicago (Central) and NYC (East Coast) ensures resilience against regional outages, data center failures, and Mullvad server maintenance windows. Different power grids and weather zones minimize correlated failures.



Status > Gateways showing both GW_VPN_CHI and GW_VPN_NYC status

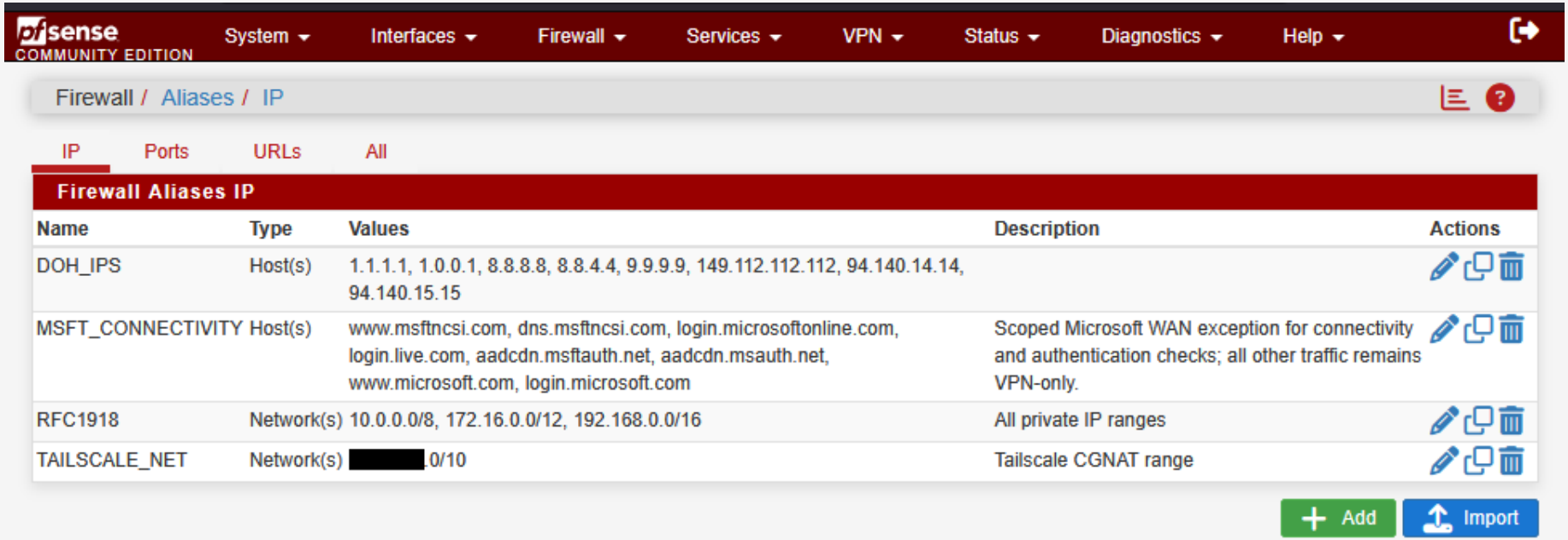


System > Routing > Gateway Groups > VPN_FAILOVER configuration

3. Firewall Aliases

The following aliases are referenced in firewall rules across all interfaces. These must be configured before creating the rules.

Alias Name	Type	Values	Purpose
DOH_IPS	Host(s)	Known DoH/DoT provider IPs (Google, Cloudflare, Quad9, etc.)	Block encrypted DNS bypass attempts
RFC1918	Network(s)	10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16	Block inter-VLAN traffic (all private IP ranges)
MSFT_CONNECTIVITY	Host(s)	Microsoft NCSI connectivity check IPs	Allow Windows network status indicator (optional)



Firewall > Aliases > IP tab showing DOH_IPS, RFC1918, and MSFT_CONNECTIVITY

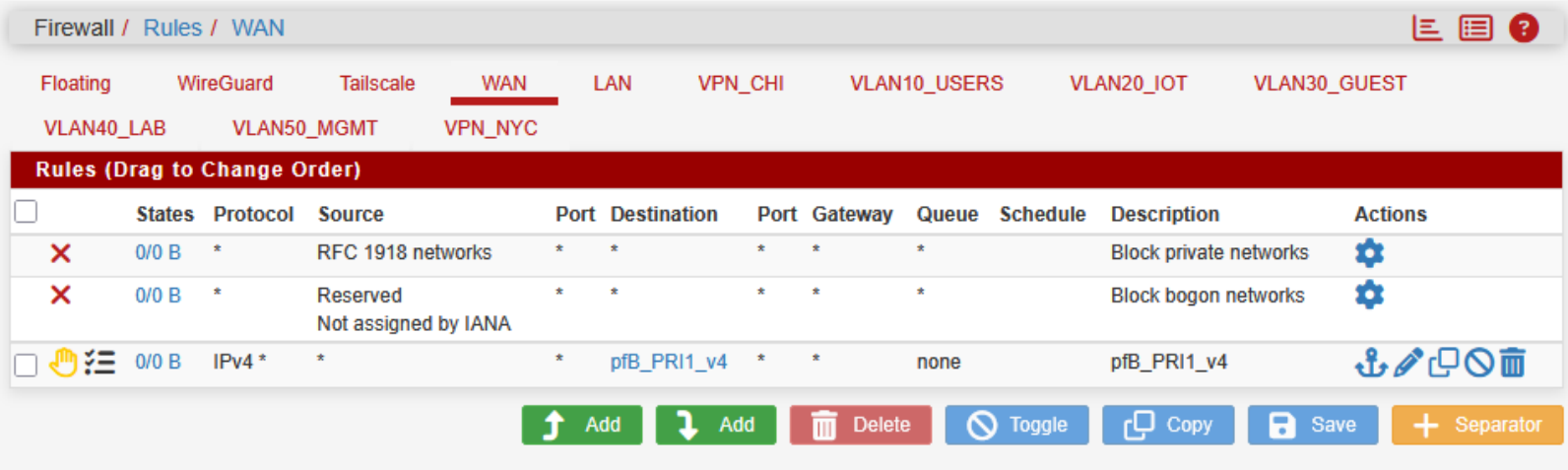
4. WAN Interface Rules

The WAN interface faces the ISP. Only inbound blocking rules exist here — no traffic is permitted inbound from the internet. pfBlockerNG adds its own rule for threat intelligence feeds.

4.1 Rule Table

#	Action	Protocol	Source	Destination	Port	Gateway	Description
1	**Block**	Any	RFC 1918 networks	*	*	none	Block private networks
2	**Block**	Any	Reserved/Not assigned by IANA	*	*	none	Block bogon networks
3	**Pass**	IPv4	*	pfB_PRI1_v4	*	none	pfBlockerNG rule

Rationale: WAN rules block all RFC1918 (private) and bogon (unallocated) source addresses from entering the network. Any inbound packet claiming to be from a private IP is spoofed and must be dropped. pfBlockerNG adds threat intelligence blocking.



Firewall >

Rules > WAN — Full rule list

5. LAN Interface Rules

The LAN interface (igb1, 10.10.1.0/24) serves as the native untagged network. It carries management traffic and provides the Anti-Lockout rule for GUI access. All internet traffic is forced through VPN_FAILOVER.

5.1 Rule Table

#	Action	Protocol	Source	Destination	Port	Gateway	Description
1	**Pass**	Any	*	LAN Address (443, 80)	443, 80	none	Anti-Lockout Rule (auto-generated)
2	**Block**	IPv4 TCP/UDP	LAN subnets	DOH_IPS	443-853	none	Block DoH/DoT — Prevent DNS bypass
3	**Block**	IPv4 TCP/UDP	LAN subnets	*	53 (DNS)	WAN_DHCP	Block DNS to WAN — Anti-leak
4	**Pass**	IPv4 TCP	LAN subnets	MSFT_CONNECTIVITY	443 (HTTPS)	WAN_DHCP	Microsoft connectivity check (optional)
5	**Pass**	IPv4	LAN subnets	LAN subnets	*	none	Allow LAN-to-LAN (no VPN)
6	**Pass**	IPv4	LAN subnets	*	*	VPN_FAILOVER	Default allow LAN to any (via VPN)
7	**Block**	IPv6	LAN subnets	*	*	none	Block all IPv6 — Prevent leaks

5.2 Rule Rationale

Rationale: Anti-Lockout is auto-generated by pfSense and cannot be disabled from the GUI. It ensures you always have access to the web interface on the LAN IP. The DoH and DNS blocks prevent devices from bypassing the local DNS resolver. The MSFT_CONNECTIVITY rule is optional — it allows Windows to show "Connected" in the system tray by reaching Microsoft's NCSI servers directly via WAN. This creates a minor VPN bypass for connectivity checks only. Remove if maximum privacy is required.

⚠ **Note:** The LAN-to-LAN pass rule (no VPN) is required for local services: DHCP, DNS resolution to pfSense, and access to devices on the same subnet. Without it, devices cannot obtain IP addresses or resolve DNS.

pfsense COMMUNITY EDITION | System | Interfaces | Firewall | Services | VPN | Status | Diagnostics | Help

Firewall / Rules / LAN

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/370 KiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
✗	0/0 B	IPv4 TCP/ UDP	LAN subnets	*	DOH_IPS	443 - 853	*	none			
✗	0/129 KiB	IPv4 TCP/ UDP	LAN subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN (anti-leak)	
✓	0/0 B	IPv4 TCP	LAN subnets	*	MSFT_ CONNECTIVITY	443 (HTTPS)	WAN_DHCP	none			
✓	0/13 KiB	IPv4 *	LAN subnets	*	LAN subnets	*	*	none		Allow LAN-to-LAN (no VPN)	
✓	0/0 B	IPv4 *	LAN subnets	*	*	*	VPN_FAILOVER	none		Default allow LAN to any rule	
✗	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none			

Add | Add | Delete | Toggle | Copy | Save | Separator

Firewall > Rules > LAN — Full rule list

6. VLAN10_USERS Rules

VLAN 10 (10.10.10.0/24) is the primary user network. Workstations and personal devices connect here via switch port 2. All traffic exits through Mullvad VPN with full inter-VLAN isolation.

6.1 Rule Table

#	Action	Protocol	Source	Destination	Port	Gateway	Description
1	**Pass**	IPv4	VLAN10_USERS subnets	VLAN10_USERS address	*	none	Allow access to pfSense gateway
2	**Pass**	IPv4 TCP/UDP	VLAN10_USERS subnets	10.10.20.5	9100	none	Allow printing to VLAN20 printer
3	**Pass**	IPv4 TCP	VLAN10_USERS subnets	10.10.1.100	80-443	none	TEMP: Switch management access (REMOVE AFTER)
4	**Block**	IPv4	VLAN10_USERS subnets	RFC1918	*	none	Block inter-VLAN — Isolate VLAN10
5	**Block**	IPv4 TCP/UDP	VLAN10_USERS subnets	*	53 (DNS)	WAN_DHCP	Block DNS to WAN — Anti-leak
6	**Block**	IPv4 TCP/UDP	VLAN10_USERS subnets	DOH_IPS	443-853	none	Block DoH/DoT — Prevent DNS bypass
7	**Pass**	IPv4	VLAN10_USERS subnets	*	*	VPN_FAILOVER	VLAN10 to Mullvad VPN
8	**Block**	IPv6	VLAN10_USERS subnets	*	*	none	Block IPv6 — Prevent leaks

6.2 Rule Rationale

Rationale: Rule order is critical. The pfSense gateway pass rule must come first to allow DNS/DHCP. The printer pass rule allows cross-VLAN printing to VLAN20 before the RFC1918 block drops all other private traffic. The TEMP switch rule provides temporary access to the GS308E management GUI during configuration — remove this rule after switch setup is finalized.

⚠ **Note:** TEMP rule for switch management (10.10.1.100) must be removed after switch configuration is complete. Leaving it creates a permanent hole in inter-VLAN isolation.

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Firewall / Rules / VLAN10_USERS

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	5/34.46 MiB	IPv4 *	VLAN10_USERS subnets	*	VLAN10_USERS address	*	*	none		Allow access to pfSense gateway	
✓	0/0 B	IPv4 TCP/UDP	VLAN10_USERS subnets	*	10.10.20.5	9100	*	none		Allow printing to VLAN20 printer	
✓	0/281 KiB	IPv4 TCP	VLAN10_USERS subnets	*	10.10.1.100	80 - 443	*	none		TEMP_VLAN10_Switch Mgmt_(REMOVE AFTER)	
✗	0/1.81 MiB	IPv4 *	VLAN10_USERS subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN10	
✗	0/64 KiB	IPv4 TCP/UDP	VLAN10_USERS subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
✗	0/0 B	IPv4 TCP/UDP	VLAN10_USERS subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
✓	12/1.94 GiB	IPv4 *	VLAN10_USERS subnets	*	*	*	VPN_FAILOVER	none		VLAN10_to_Mullvad	
✗	0/0 B	IPv6 *	VLAN10_USERS subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

Add | Add | Delete | Toggle | Copy | Save | Separator

Firewall > Rules > VLAN10_USERS — Full rule list

7. VLAN20_IOT Rules

VLAN 20 (10.10.20.0/24) hosts IoT devices and the network printer. This network is fully isolated from all other VLANs. Devices here can only reach the internet through VPN and their own pfSense gateway for DNS.

7.1 Rule Table

#	Action	Protocol	Source	Destination	Port	Gateway	Description
1	**Pass**	IPv4	VLAN20_IOT subnets	VLAN20_IOT address	*	none	Allow access to pfSense gateway
2	**Block**	IPv4 TCP/UDP	VLAN20_IOT subnets	*	53 (DNS)	WAN_DHCP	Block DNS to WAN — Anti-leak
3	**Block**	IPv4 TCP/UDP	VLAN20_IOT subnets	DOH_IPS	443-853	none	Block DoH/DoT — Prevent DNS bypass
4	**Block**	IPv4	VLAN20_IOT subnets	RFC1918	*	none	Block inter-VLAN — Isolate VLAN20
5	**Pass**	IPv4	VLAN20_IOT subnets	*	*	VPN_FAILOVER	VLAN20 to Mullvad VPN
6	**Block**	IPv6	VLAN20_IOT subnets	*	*	none	Block IPv6 — Prevent leaks

7.2 Static IP Assignments

Device	IP Address	MAC	Notes
Network Printer	10.10.20.5	[See DHCP Static Mappings]	DHCP reservation; below DHCP range

Rationale: IoT devices are inherently less secure and should be isolated. The RFC1918 block prevents any compromised IoT device from scanning or attacking devices on other VLANs. The printer is placed here for isolation but accessible from VLAN10 via a targeted pass rule (port 9100 to 10.10.20.5 only).

Firewall / Rules / VLAN20_IOT

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
✓	0/0 B	IPv4 *	VLAN20_IOT subnets	*	VLAN20_IOT address	*	*	none		Allow access to pfSense gateway	
✗	0/2 KiB	IPv4 TCP/UDP	VLAN20_IOT subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
✗	0/0 B	IPv4 TCP/UDP	VLAN20_IOT subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
✗	0/0 B	IPv4 *	VLAN20_IOT subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN20	
✓	0/0 B	IPv4 *	VLAN20_IOT subnets	*	*	*	VPN_FAILOVER	none		VLAN20_to_Mullvad	
✗	0/0 B	IPv6 *	VLAN20_IOT subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

Add | Add | Delete | Toggle | Copy | Save | Separator

Firewall > Rules > VLAN20_IOT — Full rule list

8. VLAN30_GUEST Rules

VLAN 30 (10.10.30.0/24) provides internet-only access for guest devices. No access to any internal resources. Intended for future use with a dedicated guest Wi-Fi SSID on the R6400 AP.

8.1 Rule Table

#	Action	Protocol	Source	Destination	Port	Gateway	Description
1	**Pass**	IPv4	VLAN30_GUEST subnets	VLAN30_GUEST address	*	none	Allow access to pfSense gateway
2	**Block**	IPv4 TCP/UDP	VLAN30_GUEST subnets	*	53 (DNS)	WAN_DHCP	Block DNS to WAN — Anti-leak
3	**Block**	IPv4 TCP/UDP	VLAN30_GUEST subnets	DOH_IPS	443-853	none	Block DoH/DoT — Prevent DNS bypass
4	**Block**	IPv4	VLAN30_GUEST subnets	RFC1918	*	none	Block inter-VLAN — Isolate VLAN30
5	**Pass**	IPv4	VLAN30_GUEST subnets	*	*	VPN_FAILOVER	VLAN30 to Mullvad VPN
6	**Block**	IPv6	VLAN30_GUEST subnets	*	*	none	Block IPv6 — Prevent leaks

Rationale: Guest network provides internet access through VPN while completely blocking access to all internal networks. The RFC1918 block ensures guests cannot discover or communicate with any device on the home network, including the printer, NAS, or management interfaces.

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Firewall / Rules / VLAN30_GUEST

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
0/0 B	IPv4 *	VLAN30_GUEST subnets	*	VLAN30_GUEST address	*	*	none		Allow access to pfSense gateway	
0/0 B	IPv4 TCP/UDP	VLAN30_GUEST subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
0/0 B	IPv4 TCP/UDP	VLAN30_GUEST subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
0/0 B	IPv4 *	VLAN30_GUEST subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN30	
0/0 B	IPv4 *	VLAN30_GUEST subnets	*	*	*	VPN_FAILOVER	none		VLAN30_to_Mullvad	
0/0 B	IPv6 *	VLAN30_GUEST subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

Add | Add | Delete | Toggle | Copy | Save | Separator

Firewall > Rules > VLAN30_GUEST — Full rule list

9. VLAN40_LAB Rules

VLAN 40 (10.10.40.0/24) is the lab environment for Cisco Catalyst 3560, Cisco 1900, and other lab equipment. Isolated from production VLANs to prevent lab experiments from affecting the main network.

9.1 Rule Table

#	Action	Protocol	Source	Destination	Port	Gateway	Description
1	Pass	IPv4	VLAN40_LAB subnets	VLAN40_LAB address	*	none	Allow access to pfSense gateway
2	Block	IPv4 TCP/UDP	VLAN40_LAB subnets	*	53 (DNS)	WAN_DHCP	Block DNS to WAN — Anti-leak
3	Block	IPv4 TCP/UDP	VLAN40_LAB subnets	DOH_IPS	443-853	none	Block DoH/DoT — Prevent DNS bypass
4	Block	IPv4	VLAN40_LAB subnets	RFC1918	*	none	Block inter-VLAN — Isolate VLAN40
5	Pass	IPv4	VLAN40_LAB subnets	*	*	VPN_FAILOVER	VLAN40 to Mullvad VPN
6	Block	IPv6	VLAN40_LAB subnets	*	*	none	Block IPv6 — Prevent leaks

Rationale: Lab equipment may run experimental configurations, vulnerable services, or intentionally misconfigured setups. Full isolation via RFC1918 block prevents any lab mishap from impacting production VLANs. VPN routing ensures lab traffic remains private.

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Firewall / Rules / VLAN40_LAB

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
0/0 B	IPv4 *	VLAN40_LAB subnets	*	VLAN40_LAB address	*	*	none		Allow access to pfSense gateway	
0/0 B	IPv4 TCP/UDP	VLAN40_LAB subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
0/0 B	IPv4 TCP/UDP	VLAN40_LAB subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
0/0 B	IPv4 *	VLAN40_LAB subnets	*	RFC1918	*	*	none		Block inter-VLAN — Isolate VLAN40	
0/0 B	IPv4 *	VLAN40_LAB subnets	*	*	*	VPN_FAILOVER	none		VLAN40_to_Mullvad	
0/0 B	IPv6 *	VLAN40_LAB subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

Add | Add | Delete | Toggle | Copy | Save | Separator

Firewall > Rules > VLAN40_LAB — Full rule list

10. VLAN50_MGMT Rules

VLAN 50 (10.10.50.0/24) is the management network. It has unrestricted access to all VLANs for administration purposes. Traffic exits directly to the internet without VPN to ensure management access even if all VPN tunnels are down.

10.1 Rule Table

#	Action	Protocol	Source	Destination	Port	Gateway	Description
1	**Pass**	IPv4	VLAN50_MGMT subnets	VLAN50_MGMT address	*	none	Allow access to pfSense gateway
2	**Block**	IPv4 TCP/UDP	VLAN50_MGMT subnets	*	53 (DNS)	WAN_DHCP	Block DNS to WAN — Anti-leak
3	**Block**	IPv4 TCP/UDP	VLAN50_MGMT subnets	DOH_IPS	443-853	none	Block DoH/DoT — Prevent DNS bypass
4	**Pass**	IPv4	VLAN50_MGMT subnets	*	*	none	VLAN50_MGMT — Direct internet (no VPN)

10.2 Key Differences from Other VLANs

- No RFC1918 block — MGMT must reach all VLANs (10.10.1.x, 10.10.10.x, 10.10.20.x, etc.) for administration
- No VPN gateway — Management traffic routes directly through WAN. If VPN tunnels fail, MGMT still has internet and full network access
- No IPv6 block — Optional, can be added for consistency

Rationale: The management VLAN is the “break glass” network. If both VPN tunnels go down and all other VLANs lose internet, a device on VLAN50 can still reach pfSense, the switch, the AP, and the internet to diagnose and resolve issues. Physical access to this VLAN should be restricted — only one switch port (port 8) is assigned.

⚠ **Note:** VLAN50_MGMT traffic is NOT encrypted by VPN. Any device on this VLAN exposes its real ISP IP address. Use only for management tasks, never for general browsing.

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Firewall / Rules / VLAN50_MGMT

Floating | WireGuard | Tailscale | WAN | LAN | VPN_CHI | VLAN10_USERS | VLAN20_IOT | VLAN30_GUEST | VLAN40_LAB | VLAN50_MGMT | VPN_NYC

Rules (Drag to Change Order)

States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
0/0 B	IPv4 *	VLAN50_MGMT subnets	*	VLAN50_MGMT address	*	*	none		Allow access to pfSense gateway	
0/0 B	IPv4 TCP/UDP	VLAN50_MGMT subnets	*	*	53 (DNS)	WAN_DHCP	none		Block DNS to WAN — Anti-leak	
0/0 B	IPv4 TCP/UDP	VLAN50_MGMT subnets	*	DOH_IPS	443 - 853	*	none		Block DoH/DoT — Prevent DNS bypas	
0/0 B	IPv4 *	VLAN50_MGMT subnets	*	*	*	*	none		VLAN50_MGMT — Direct internet (no VPN)	
0/0 B	IPv6 *	VLAN50_MGMT subnets	*	*	*	*	none		Block IPv6 — Prevent leaks	

Add | Add | Delete | Toggle | Copy | Save | Separator

Firewall > Rules > VLAN50_MGMT — Full rule list

11. Outbound NAT Rules

Manual Outbound NAT is configured to translate traffic from each subnet through the correct VPN interface. Each subnet has two NAT rules — one for VPN_CHI and one for VPN_NYC to support failover.

11.1 VPN_CHI NAT Rules

Interface	Source	NAT Address	Description
VPN_CHI	10.10.1.0/24	VPN_CHI address	LAN to VPN_CHI
VPN_CHI	10.10.10.0/24	VPN_CHI address	VLAN10_USERS to VPN_CHI
VPN_CHI	10.10.20.0/24	VPN_CHI address	VLAN20_IOT to VPN_CHI
VPN_CHI	10.10.30.0/24	VPN_CHI address	VLAN30_GUEST to VPN_CHI
VPN_CHI	10.10.40.0/24	VPN_CHI address	VLAN40_LAB to VPN_CHI
VPN_CHI	10.10.50.0/24	VPN_CHI address	VLAN50_MGMT to VPN_CHI

11.2 VPN_NYC NAT Rules

Interface	Source	NAT Address	Description
VPN_NYC	10.10.1.0/24	VPN_NYC address	LAN to VPN_NYC
VPN_NYC	10.10.10.0/24	VPN_NYC address	VLAN10_USERS to VPN_NYC
VPN_NYC	10.10.20.0/24	VPN_NYC address	VLAN20_IOT to VPN_NYC
VPN_NYC	10.10.30.0/24	VPN_NYC address	VLAN30_GUEST to VPN_NYC
VPN_NYC	10.10.40.0/24	VPN_NYC address	VLAN40_LAB to VPN_NYC
VPN_NYC	10.10.50.0/24	VPN_NYC address	VLAN50_MGMT to VPN_NYC

Rationale: Without NAT rules on the VPN interface, traffic routed through the tunnel will be dropped at the exit node. Both VPN interfaces need NAT rules for every subnet to support seamless failover. When GW_VPN_CHI fails, traffic shifts to GW_VPN_NYC, which must also have NAT rules to translate the source address.

⚠ **Note:** No WAN NAT rules exist for internal subnets. This is intentional — the kill switch design. If VPN is down, traffic cannot fall back to WAN because there are no NAT rules to translate it.

Firewall / NAT / Outbound

Port Forward | 1:1 | Outbound | NPt

Outbound NAT Mode

Mode			
	○ Automatic outbound NAT rule generation. (IPsec passthrough included)	○ Hybrid Outbound NAT rule generation. (Automatic Outbound NAT + rules below)	◉ Manual Outbound NAT rule generation. (AON - Advanced Outbound NAT)	○ Disable Outbound NAT rule generation. (No Outbound NAT rules)

Save

Mappings

Interface	Source	Source Port	Destination	Destination Port	NAT Address	NAT Port	Static Port	Description	Actions
VPN_CHI	10.10.1.0/24	*	*	*	VPN_CHI address	*		LAN to VPN_CHI	
VPN_CHI	10.10.10.0/24	*	*	*	VPN_CHI address	*		VLAN10_USERS to VPN_CHI	
VPN_CHI	10.10.20.0/24	*	*	*	VPN_CHI address	*		VLAN20_IOT to VPN_CHI	
VPN_CHI	10.10.30.0/24	*	*	*	VPN_CHI address	*		VLAN30_GUEST to VPN_CHI	
VPN_CHI	10.10.40.0/24	*	*	*	VPN_CHI address	*		VLAN40_LAB to VPN_CHI	
VPN_CHI	10.10.50.0/24	*	*	*	VPN_CHI address	*		VLAN50_MGMT to VPN_CHI	
VPN_NYC	10.10.1.0/24	*	*	*	VPN_NYC address	*		LAN to VPN_NYC	
VPN_NYC	10.10.10.0/24	*	*	*	VPN_NYC address	*		VLAN10_USERS to VPN_NYC	
VPN_NYC	10.10.20.0/24	*	*	*	VPN_NYC address	*		VLAN20_IOT to VPN_NYC	
VPN_NYC	10.10.30.0/24	*	*	*	VPN_NYC address	*		VLAN30_GUEST to VPN_NYC	
VPN_NYC	10.10.40.0/24	*	*	*	VPN_NYC address	*		VLAN40_LAB to VPN_NYC	
VPN_NYC	10.10.50.0/24	*	*	*	VPN_NYC address	*		VLAN50_MGMT to VPN_NYC	

Add | Add | Delete | Toggle | Save

Firewall > NAT > Outbound — Full mapping list showing all 12 rules

12. VPN Kill Switch Design

The kill switch prevents any traffic from reaching the internet if both VPN tunnels are offline. It is implemented through multiple layers:

Layer 1: Gateway Routing

All VLAN firewall pass rules use VPN_FAILOVER as the gateway. If both GW_VPN_CHI and GW_VPN_NYC are offline, the gateway group has no active members and traffic is dropped — it never falls back to WAN.

Layer 2: Outbound NAT

NAT rules exist only on VPN_CHI and VPN_NYC interfaces. No WAN NAT rules for internal subnets means even if traffic somehow reaches WAN, it cannot be translated and will be dropped.

Layer 3: DNS Lockdown

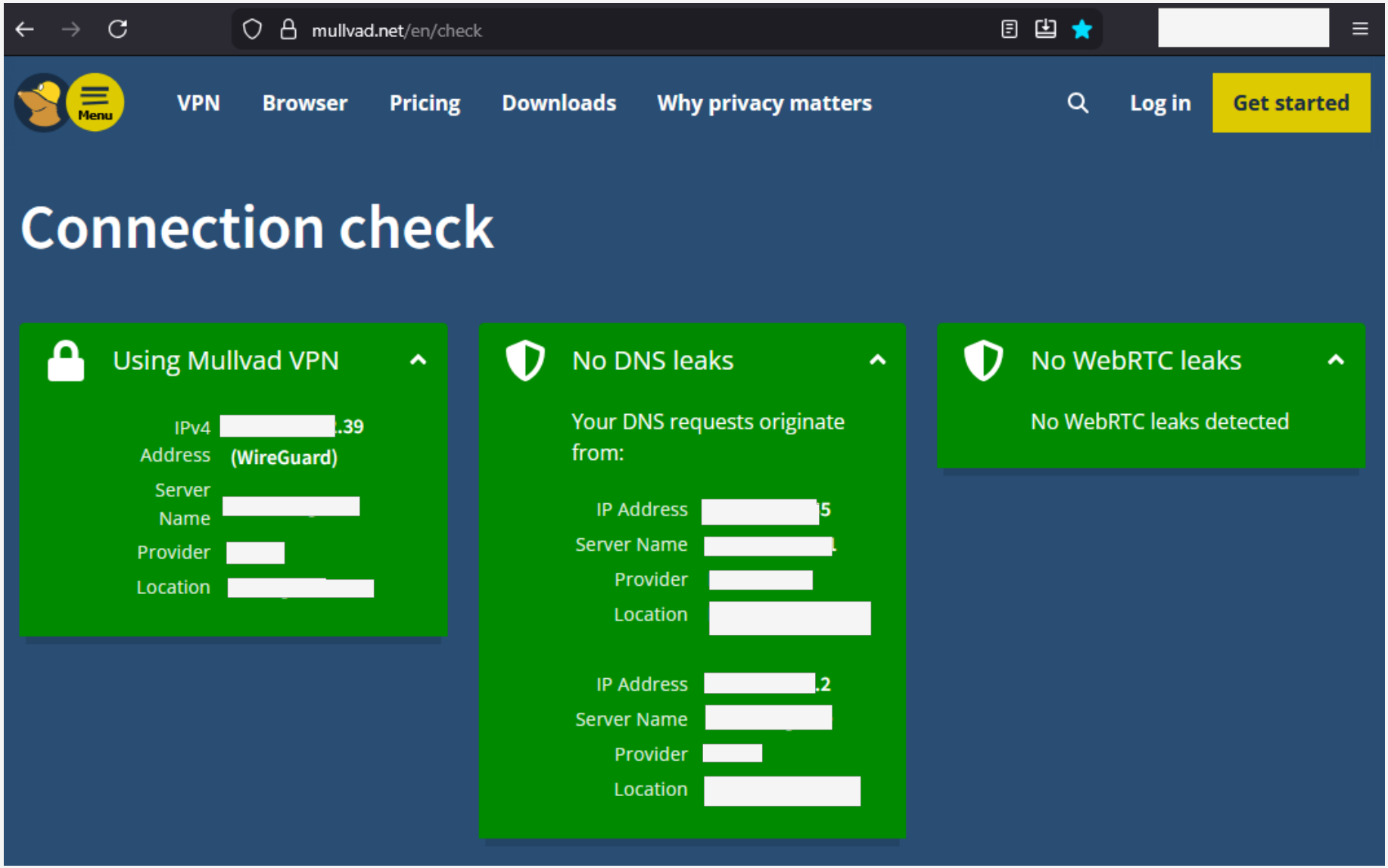
DNS Resolver outgoing interfaces are restricted to VPN_CHI and VPN_NYC. System DNS servers are split between GW_VPN_CHI and GW_VPN_NYC gateways. If VPN is down, DNS resolution fails, preventing any domain-based connection attempts via WAN.

Layer 4: IPv6 Blocking

IPv6 is blocked on every interface. This prevents any IPv6 traffic from bypassing the IPv4 VPN tunnel, which is a common VPN leak vector.

Layer 5: DNS Bypass Prevention

DoH (DNS over HTTPS) and DoT (DNS over TLS) are blocked via the DOH_IPS alias on ports 443-853. Direct DNS to the ISP is blocked via port 53 to WAN_DHCP gateway. Devices must use the local pfSense resolver, which only queries through VPN.



Kill switch verification: ipleak.net results showing Mullvad VPN, no DNS leaks, no WebRTC leaks

13. Future Planning

13.1 Planned Changes

- **Disable TEMP switch management rule** from VLAN10_USERS after switch configuration is finalized
- Evaluate MSFT_CONNECTIVITY rule on LAN — remove if maximum privacy is desired
- Configure guest Wi-Fi SSID on R6400 AP mapped to VLAN30_GUEST – Done but have limitation (AP mode support no VLAN supporting device)
- Add additional printer ports (443, 631/IPP) to VLAN10 printer rule if needed - NA
- Add IPv6 block to VLAN50_MGMT for consistency - Done

13.2 Potential Enhancements

- Rate limiting on VLAN30_GUEST to prevent bandwidth abuse by guest devices
- Captive portal for VLAN30_GUEST requiring acceptance of terms before internet access
- IDS/IPS via Suricata or Snort on VPN interfaces for threat detection
- Additional VPN endpoints (West Coast, international) for multi-region failover
- Scheduled rules — time-based restrictions on VLAN20_IOT (e.g., IoT devices only allowed internet during specific hours)
- Dedicated VLAN for Tailscale remote access traffic
- VLAN40_LAB inter-VLAN exception — allow specific lab-to-lab traffic if Cisco gear needs to communicate with production monitoring

13.3 Rule Change Procedure

1. Document the proposed change in this document before implementation
2. Take a pfSense configuration backup (Diagnostics > Backup & Restore)
3. Implement the change on a single interface first
4. Test connectivity from a device on that VLAN
5. Run leak tests (ipleak.net, mullvad.net/check) to verify kill switch integrity
6. Update this document with the final rule configuration and screenshot
7. Take a post-change backup